

1. Write a shell script which will execute the following set of tasks:

```
vivswsyvora@MacBook-Air-de-VivswSyvora ~ % ls
Desktop      Documents    Downloads    Library      Movies      Music      Pictures    Public
vivswsyvora@MacBook-Air-de-VivswSyvora ~ % mkdir -p ~/sample
vivswsyvora@MacBook-Air-de-VivswSyvora ~ % ls
Desktop      Documents    Downloads    Library      Movies      Music      Pictures    Public      sample
vivswsyvora@MacBook-Air-de-VivswSyvora ~ % cd sample
vivswsyvora@MacBook-Air-de-VivswSyvora sample % touch sample.txt
vivswsyvora@MacBook-Air-de-VivswSyvora sample % ls
sample.txt
vivswsyvora@MacBook-Air-de-VivswSyvora sample % echo "Hi! This is a sample text file created using shell script." > sample.txt
vivswsyvora@MacBook-Air-de-VivswSyvora sample % cat sample.txt
Hi! This is a sample text file created using shell script.
vivswsyvora@MacBook-Air-de-VivswSyvora sample % grep -o "t" sample.txt | wc -l
4
vivswsyvora@MacBook-Air-de-VivswSyvora sample % chmod u+rx sample.txt
vivswsyvora@MacBook-Air-de-VivswSyvora sample % echo " Hi! This is just another sample text added to the file." >> sample.txt
vivswsyvora@MacBook-Air-de-VivswSyvora sample % cat sample.txt
Hi! This is a sample text file created using shell script.
Hi! This is just another sample text added to the file.
vivswsyvora@MacBook-Air-de-VivswSyvora sample % chmod g=r sample.txt
vivswsyvora@MacBook-Air-de-VivswSyvora sample % chmod o= sample.txt
vivswsyvora@MacBook-Air-de-VivswSyvora sample % cp sample.txt sample2.txt
vivswsyvora@MacBook-Air-de-VivswSyvora sample % ls
sample.txt      sample2.txt
vivswsyvora@MacBook-Air-de-VivswSyvora sample % cat sample2.txt
Hi! This is a sample text file created using shell script.
Hi! This is just another sample text added to the file.
```

```
vivswsyvora@MacBook-Air-de-VivswSyvora sample % for i in {1..1000}; do echo "Random Line $i" >> sample.txt; done
vivswsyvora@MacBook-Air-de-VivswSyvora sample % cat sample.txt
Hi! This is a sample text file created using shell script.
Hi! This is just another sample text added to the file.
Random Line 1
Random Line 2
Random Line 3
Random Line 4
Random Line 5
Random Line 6
Random Line 7
Random Line 8
Random Line 9
Random Line 10
Random Line 11
Random Line 12
Random Line 13
Random Line 14
Random Line 15
Random Line 16
Random Line 17
Random Line 18
Random Line 19
Random Line 20
Random Line 21
Random Line 22
Random Line 23
Random Line 24
Random Line 25
Random Line 26
Random Line 27
Random Line 28
Random Line 29
Random Line 30
Random Line 31
Random Line 32
Random Line 33
Random Line 34
Random Line 35
Random Line 36
Random Line 37
Random Line 38
Random Line 39
Random Line 40
Random Line 41
Random Line 42
Random Line 43
Random Line 44
Random Line 45
Random Line 46
Random Line 47
Random Line 48
Random Line 49
Random Line 50
Random Line 51
Random Line 52
Random Line 53
Random Line 54
Random Line 55
Random Line 56
Random Line 57
Random Line 58
Random Line 59
Random Line 60
```

Random Line 937
Random Line 938
Random Line 939
Random Line 940
Random Line 941
Random Line 942
Random Line 943
Random Line 944
Random Line 945
Random Line 946
Random Line 947
Random Line 948
Random Line 949
Random Line 950
Random Line 951
Random Line 952
Random Line 953
Random Line 954
Random Line 955
Random Line 956
Random Line 957
Random Line 958
Random Line 959
Random Line 960
Random Line 961
Random Line 962
Random Line 963
Random Line 964
Random Line 965
Random Line 966
Random Line 967
Random Line 968
Random Line 969
Random Line 970
Random Line 971
Random Line 972
Random Line 973
Random Line 974
Random Line 975
Random Line 976
Random Line 977
Random Line 978
Random Line 979
Random Line 980
Random Line 981
Random Line 982
Random Line 983
Random Line 984
Random Line 985
Random Line 986
Random Line 987
Random Line 988
Random Line 989
Random Line 990
Random Line 991
Random Line 992
Random Line 993
Random Line 994
Random Line 995
Random Line 996
Random Line 997
Random Line 998
Random Line 999
Random Line 1000
vivswsyvora@MacBoo

```
vivswsyvora@MacBook-Air-de-VivswSyvora sample % head -n 50 sample.txt
Hi! This is a sample text file created using shell script.
Hi! This is just another sample text added to the file.
Random Line 1
Random Line 2
Random Line 3
Random Line 4
Random Line 5
Random Line 6
Random Line 7
Random Line 8
Random Line 9
Random Line 10
Random Line 11
Random Line 12
Random Line 13
Random Line 14
Random Line 15
Random Line 16
Random Line 17
Random Line 18
Random Line 19
Random Line 20
Random Line 21
Random Line 22
Random Line 23
Random Line 24
Random Line 25
Random Line 26
Random Line 27
Random Line 28
Random Line 29
Random Line 30
Random Line 31
Random Line 32
Random Line 33
Random Line 34
Random Line 35
Random Line 36
Random Line 37
Random Line 38
Random Line 39
Random Line 40
Random Line 41
Random Line 42
Random Line 43
Random Line 44
Random Line 45
Random Line 46
Random Line 47
Random Line 48
vivswsyvora@MacBook-Air-de-VivswSyvora sample %
```

```
vivswsyvora@MacBook-Air-de-VivswSyvora sample % tail -n 50 sample.txt
Random Line 951
Random Line 952
Random Line 953
Random Line 954
Random Line 955
Random Line 956
Random Line 957
Random Line 958
Random Line 959
Random Line 960
Random Line 961
Random Line 962
Random Line 963
Random Line 964
Random Line 965
Random Line 966
Random Line 967
Random Line 968
Random Line 969
Random Line 970
Random Line 971
Random Line 972
Random Line 973
Random Line 974
Random Line 975
Random Line 976
Random Line 977
Random Line 978
Random Line 979
Random Line 980
Random Line 981
Random Line 982
Random Line 983
Random Line 984
Random Line 985
Random Line 986
Random Line 987
Random Line 988
Random Line 989
Random Line 990
Random Line 991
Random Line 992
Random Line 993
Random Line 994
Random Line 995
Random Line 996
Random Line 997
Random Line 998
Random Line 999
Random Line 1000
```

```

vivswsyvora@MacBook-Air-de-VivswSyvora sample % ls
sample.txt      sample2.txt
vivswsyvora@MacBook-Air-de-VivswSyvora sample % touch prog1.txt prog2.txt program.txt code.txt info.txt
vivswsyvora@MacBook-Air-de-VivswSyvora sample % ls
code.txt      info.txt      prog1.txt      prog2.txt      program.txt      sample.txt      sample2.txt
vivswsyvora@MacBook-Air-de-VivswSyvora sample % ls grep "prog"
ls: grep: No such file or directory
ls: prog: No such file or directory
vivswsyvora@MacBook-Air-de-VivswSyvora sample % ls *prog*
prog1.txt      prog2.txt      program.txt
vivswsyvora@MacBook-Air-de-VivswSyvora sample % ls ~/sample/*prog*
/Users/vivswsyvora/sample/prog1.txt  /Users/vivswsyvora/sample/prog2.txt  /Users/vivswsyvora/sample/program.txt
vivswsyvora@MacBook-Air-de-VivswSyvora sample % echo "alias 'list_prog'='ls ~/sample/*prog*' " >> ~/.bashrc
vivswsyvora@MacBook-Air-de-VivswSyvora sample % alias 'list_prog'
vivswsyvora@MacBook-Air-de-VivswSyvora sample % source ~/.bashrc
vivswsyvora@MacBook-Air-de-VivswSyvora sample % alias 'list_prog'
list_prog='ls ~/sample/*prog*'
vivswsyvora@MacBook-Air-de-VivswSyvora sample % list_prog
/Users/vivswsyvora/sample/prog1.txt  /Users/vivswsyvora/sample/prog2.txt  /Users/vivswsyvora/sample/program.txt
vivswsyvora@MacBook-Air-de-VivswSyvora sample % cd ..
vivswsyvora@MacBook-Air-de-VivswSyvora ~ % list_prog
zsh: command not found: list
vivswsyvora@MacBook-Air-de-VivswSyvora ~ % list_prog
/Users/vivswsyvora/sample/prog1.txt  /Users/vivswsyvora/sample/prog2.txt  /Users/vivswsyvora/sample/program.txt
vivswsyvora@MacBook-Air-de-VivswSyvora ~ %
vivswsyvora@MacBook-Air-de-VivswSyvora ~ % ls ~/sample/*prog*
/Users/vivswsyvora/sample/prog1.txt  /Users/vivswsyvora/sample/prog2.txt  /Users/vivswsyvora/sample/program.txt
vivswsyvora@MacBook-Air-de-VivswSyvora ~ % list(){
function> ls ~/sample/*"$1"*
function> }
vivswsyvora@MacBook-Air-de-VivswSyvora ~ % source ~/.bashrc
zsh: no such user or named directory: .
vivswsyvora@MacBook-Air-de-VivswSyvora ~ % source ~/.bashrc
vivswsyvora@MacBook-Air-de-VivswSyvora ~ % list_prog
/Users/vivswsyvora/sample/prog1.txt  /Users/vivswsyvora/sample/prog2.txt  /Users/vivswsyvora/sample/program.txt
vivswsyvora@MacBook-Air-de-VivswSyvora ~ %

```

2. What is the difference between the source and sh commands?

When you call source or . you load and execute a shell script into the current shell process. Changes like setting variables, aliases etc persist in current session. For eg. source ~/.bashrc

When you call sh, you initiate a fork (sub-process or child) that runs a new session of /bin/sh (which is often a symbolic link to bash). In this case, environment variables set by the sub-script would be dropped when the sub-script terminate i.e. sh runs a script in new shell process so changes don't affect current session. For eg. sh script.sh

This means that while source modifies current shell sh doesn't do that.

```

vivswsyvora@192 sample % echo 'export XYZ_VAR="Hi Scripter"' > sampletest.sh
vivswsyvora@192 sample % sh test.sh
sh: test.sh: No such file or directory
vivswsyvora@192 sample % sh sampletest.sh
vivswsyvora@192 sample % echo $XYZ_VAR

vivswsyvora@192 sample % source sampletest.sh
vivswsyvora@192 sample % echo $XYZ_VAR
Hi Scripter

```



```

vivswsyvora@192 ~ % cd sample
vivswsyvora@192 sample % echo 'export XYZ_VAR="Hi Scriptor"' > sampletest.sh
vivswsyvora@192 sample % sh test.sh
sh: test.sh: No such file or directory
vivswsyvora@192 sample % sh sampletest.sh
vivswsyvora@192 sample % echo $XYZ_VAR
vivswsyvora@192 sample % source sampletest.sh
vivswsyvora@192 sample % echo $XYZ_VAR
Hi Scriptor
vivswsyvora@192 sample %

```

3. Create two files “a.txt” and “b.txt”. Write a command to get the difference between the contents in two files.

```

[vivswsyvora@MacBook-Air-de-VivswSyvora ~ % cd sample
[vivswsyvora@MacBook-Air-de-VivswSyvora sample % echo "ABC
[quote> PQR
[quote> XYZ" > a.txt
vivswsyvora@MacBook-Air-de-VivswSyvora sample % echo "ABC
MNO
[XYZ" > b.txt
[vivswsyvora@MacBook-Air-de-VivswSyvora sample % diff a.txt b.txt
2c2
< PQR
---
> MNO
[vivswsyvora@MacBook-Air-de-VivswSyvora sample % cat a.txt
ABC
PQR
XYZ
[vivswsyvora@MacBook-Air-de-VivswSyvora sample % cat b.txt
ABC
MNO
XYZ
vivswsyvora@MacBook-Air-de-VivswSyvora sample % █

```

```

vivswsyvora@192 sample % cat a.txt
ABC
PQR
XYZ
vivswsyvora@192 sample % cat b.txt
ABC
MNO
XYZ
vivswsyvora@192 sample % diff a.txt b.txt
2c2
< PQR
---
> MNO

```

4. What is the difference between ls and ls -l?

We can use the ls command to list files and directories in any given or current directory. For eg.

```
[vivswsyvora@MacBook-Air-de-VivswSyvora ~ % ls
Desktop      Downloads    Movies       Pictures     sample
Documents    Library      Music        Public
[vivswsyvora@MacBook-Air-de-VivswSyvora ~ % ls ~/sample
a.txt        code.txt     prog1.txt    program.txt  sample2.txt
b.txt        info.txt     prog2.txt    sample.txt
[vivswsyvora@MacBook-Air-de-VivswSyvora ~ % cd sample
[vivswsyvora@MacBook-Air-de-VivswSyvora sample % ls
a.txt        code.txt     prog1.txt    program.txt  sample2.txt
b.txt        info.txt     prog2.txt    sample.txt
vivswsyvora@MacBook-Air-de-VivswSyvora sample % █
```

On the other hand lsof lists open files and shows which processes are using them. Imagine lsof as a ls with the addition of "open files". For eg.

```
[vivswsyvora@192 ~ % ls
Desktop      Documents    Downloads     Library       Movies
Music        Pictures     Public        sample
[vivswsyvora@192 ~ % lsof /etc/hosts
COMMAND  PID    USER    FD  TYPE DEVICE SIZE/OFF  NODE NAME
Google   873    vivswsyvora 23r REG  1,14   213 16269 /private/etc/hosts
Google   883    vivswsyvora 16r REG  1,14   213 16269 /private/etc/hosts
vivswsyvora@192 ~ % █
```

```
vivswsyvora@192 ~ % ls
Desktop      Downloads    Movies       Pictures hello
Documents    Library      Music        Public    sample
vivswsyvora@192 ~ % ls ~/sample
a.txt        code.txt  prog1.txt  program.txt  sample2.txt
b.txt        info.txt  prog2.txt  sample.txt   sampletest.sh
vivswsyvora@192 ~ % cd sample
vivswsyvora@192 sample % ls
a.txt        code.txt  prog1.txt  program.txt  sample2.txt
b.txt        info.txt  prog2.txt  sample.txt   sampletest.sh
vivswsyvora@192 sample % cd ..
vivswsyvora@192 ~ % ls
Desktop      Downloads    Movies       Pictures hello
Documents    Library      Music        Public    sample
vivswsyvora@192 ~ % lsof /etc/hosts
COMMAND  PID    USER    FD  TYPE DEVICE SIZE/OFF  NODE NAME
Google   873    vivswsyvora 23r REG  1,14   213 16269 /private/etc/hosts
Google   883    vivswsyvora 16r REG  1,14   213 16269 /private/etc/hosts
```

5. Create directories ./hello/world (World dir is inside hello dir) using mkdir command where neither hello or world exists. It should be a single command without the use of &&.

```

vivswsyvora@192 ~ % ls
Desktop      Downloads    Movies       Pictures     sample
Documents    Library      Music        Public
vivswsyvora@192 ~ % mkdir -p ./hello/world
vivswsyvora@192 ~ % ls
Desktop      Downloads    Movies       Pictures     hello
Documents    Library      Music        Public       sample
vivswsyvora@192 ~ % cd hello
vivswsyvora@192 hello % ls
world
vivswsyvora@192 hello % cd world
vivswsyvora@192 world % █

```

The -p flag helps create the required nested directories otherwise we would get an error as the hello folder doesn't exist already. The -p flag creates the required levels of parent directories as needed if they don't exist.

```

vivswsyvora@192 ~ % ls
Desktop      Downloads    Movies       Pictures     sample
Documents    Library      Music        Public
vivswsyvora@192 ~ % mkdir -p ./hello/world
vivswsyvora@192 ~ % ls
Desktop      Downloads    Movies       Pictures     hello
Documents    Library      Music        Public       sample
vivswsyvora@192 ~ % cd hello
vivswsyvora@192 hello % ls
world
vivswsyvora@192 hello % cd world
vivswsyvora@192 world % █

```

6. How can you permanently set an environment variable using a bash terminal?

```

vivswsyvora@192 ~ % echo "export BIG_PPPABXYZ_VAR='Hey World'" >> ~/.bashrc
vivswsyvora@192 ~ % source ~/.bashrc
vivswsyvora@192 ~ % echo "Set global variable is $BIG_PPPABXYZ_VAR"
Set global variable is Hey World
vivswsyvora@192 ~ % █

```

```

vivswsyvora@192 ~ % echo "export BIG_PPPABXYZ_VAR='Hey World'" >> ~/.bashrc
vivswsyvora@192 ~ % source ~/.bashrc
vivswsyvora@192 ~ % echo "Set global variable is $BIG_PPPABXYZ_VAR"
Set global variable is Hey World
vivswsyvora@192 ~ % █

```

The .bashrc file is a configuration file for the Bash shell. The file consists of commands, functions, aliases, and scripts that execute every time a Bash session starts on Linux or macOS. The file allows customizing the shell environment with various functionalities, shortcuts, and visual tweaks.

7. You have some process running on a port in your system. How can you view and then kill the process from the terminal?

```
[vivswsyvora@192 ~ % nc -l 13254 &
[1] 1482
[vivswsyvora@192 ~ % lsof -i :13254
COMMAND PID      USER   FD   TYPE             DEVICE SIZE/OFF NODE NAME
nc       1482 vivswsyvora   3u  IPv4 0x5f1788627bc067b3      0t0  TCP *:13254 (LISTEN)
[vivswsyvora@192 ~ % kill -9 1482
vivswsyvora@192 ~ %
[1]  + killed      nc -l 13254
[vivswsyvora@192 ~ % lsof -i :13254
vivswsyvora@192 ~ %
```

nc -l 13254 starts a netcat (nc) listener (-l) on port 13254 in background (&) which we then kill using the kill command with help of process id (PID) obtained from lsof command. The -9 in kill -9 helps to immediately terminate the process, it is the kill signal (SIGKILL).

```
vivswsyvora@192 ~ % nc -l 13254 &
[1] 1482
vivswsyvora@192 ~ % lsof -i :13254
COMMAND PID      USER   FD   TYPE             DEVICE SIZE/OFF NODE NAME
nc       1482 vivswsyvora   3u  IPv4 0x5f1788627bc067b3      0t0  TCP *:13254 (LISTEN)
vivswsyvora@192 ~ % kill -9 1482
vivswsyvora@192 ~ %
[1]  + killed      nc -l 13254
vivswsyvora@192 ~ % lsof -i :13254
vivswsyvora@192 ~ %
```

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